



10 CFR 50.73

LG-17-070
May 22, 2017

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 1
Renewed Facility Operating License No. NPF-39
NRC Docket Nos. 50-352

Subject: LER 2017-003-00, Condition Prohibited by TS Due to Inoperable RPIS

Enclosed is a Licensee Event Report (LER) which addresses a condition prohibited by Technical Specifications (TS) at Limerick Generating Station (LGS) Unit 1. The condition occurred when a Rod Position Indication System (RPIS) power supply failed. The alternate indication used to satisfy the TS action was subsequently determined to be invalid, given the failed power supply.

The failure to meet the TS Action constituted a condition prohibited by TS. Therefore, this LER is being submitted pursuant to the requirements of 50.73(a)(2)(i)(B) for an Operation or Condition Prohibited by TS.

There are no commitments contained in this letter.

If you have any questions, please contact Robert B. Dickinson at (610) 718-3400.

Respectfully,

A handwritten signature in black ink, appearing to read "R. Libra".

Richard W. Libra
Vice President – Limerick Generating Station
Exelon Generation Company, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, LGS

NRC FORM 366 (04-2017)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 03/31/2020							
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block)											
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.									
1. FACILITY NAME Limerick Generating Station, Unit 1				2. DOCKET NUMBER 05000352		3. PAGE 1 OF 4							
4. TITLE Condition Prohibited by TS Due to Inoperable RPIS													
5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED				
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER			
03	22	17	2017	- 003	- 00	05	22	17	FACILITY NAME	DOCKET NUMBER			
9. OPERATING MODE		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)											
1		☐ 20.2201(b)				☐ 20.2203(a)(3)(i)				☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
		☐ 20.2201(d)				☐ 20.2203(a)(3)(ii)				☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
		☐ 20.2203(a)(1)				☐ 20.2203(a)(4)				☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
		☐ 20.2203(a)(2)(i)				☐ 50.36(c)(1)(i)(A)				☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
100		☐ 20.2203(a)(2)(ii)				☐ 50.36(c)(1)(ii)(A)				☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
		☐ 20.2203(a)(2)(iii)				☐ 50.36(c)(2)				☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
		☐ 20.2203(a)(2)(iv)				☐ 50.46(a)(3)(ii)				☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
		☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(A)				☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
		☐ 20.2203(a)(2)(vi)				☒ 50.73(a)(2)(i)(B)				☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
						☐ 50.73(a)(2)(i)(C)				☐ OTHER Specify in Abstract below or in NRC Form 366A			
12. LICENSEE CONTACT FOR THIS LER													
LICENSEE CONTACT Robert B. Dickinson, Manager – Regulatory Assurance								TELEPHONE NUMBER (Include Area Code) (610) 718-3400					
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT													
CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX				
N/A	N/A	NA	N/A	N/A	N/A	N/A	N/A	N/A	N/A				
14. SUPPLEMENTAL REPORT EXPECTED								15. EXPECTED SUBMISSION DATE					
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO								MONTH	DAY	YEAR			
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines) A power supply in the Rod Position Indication System (RPIS) failed causing a Main Control Room (MCR) alarm and loss of several control rod position indications. Using the Four Rod Display, the control room operators determined the affected control rods and found approximately half of the control rods had no indication. Although the Four Rod Display did not show control rod position for these rods, the staff concluded that the Full Core Display indication satisfied control rod position indication requirements for a majority of the control rods that were either full-in or full-out. It was later determined that the Full Core Display indicates the last known position in memory which is not replaced until new data is received. In this case, no data would have been received for the control rods due to the power supply failure. The failure to identify loss of control rod position indication for the control rods caused a failure to meet the plant's Technical Specifications (TS) and is reportable under 10 CFR 50.73(a)(2)(i)(B). This condition existed for approximately 19.5 hours prior to the power supply replacement, which was greater than the TS required actions to determine rod position within 1 hour or be in hot shutdown within the following 12 hours.													

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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		YEAR	SEQUENTIAL NUMBER	REV NO.
Limerick Generating Station, Unit 1	05000352	2017	- 003	- 00

NARRATIVE**I. Unit Conditions Prior to the Event**

Limerick Generating Station (LGS) Unit 1 was operating in Operational Condition (OPCON) 1 at 100 percent power at the time of the event. There were no other structures, systems, or components inoperable at the time of the event that contributed to the event.

II. Description of the Event

On 3/16/17 at 1724, the Unit 1 Main Control Room (MCR) received a Rod Position Indication System (RPIS) Inoperable alarm and identified that multiple control rod position indications were inoperable due to a failed power supply. The crew entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.1.3.7 Action a.1 which directs, "with one or more control rod position indicators inoperable, within 1 hour, determine the position of the control rod by using an alternate method." Of the 83 control rods with no indication on the four rod display, 81 control rods indicated the expected position of full-in or full-out on the full core display. A method of alternate control rod position determination was initiated for the remaining two control rods that were mid-position and thus had no indication.

On 3/16/17 at 1819, to satisfy the TS LCO Action, the crew determined control rod position of the two control rods by using the RPIS test box. Therefore, the TS LCO 3.1.3.7 Action was documented as being satisfied.

On 3/17/17 at 1251, the power supply was replaced and RPIS position indication was restored to operable. LCO 3.1.3.7 action was exited for all control rods.

On 3/22/17, it was identified that the power supply failure rendered the full core display incapable of updating in response to a position change for the affected control rods. Subsequently, it was determined to not be a valid means that could be used to verify control rod position indication.

The condition existed for approximately 19.5 hours prior to the power supply replacement, which is greater than the TS required actions to determine rod position within 1 hour or be in hot shutdown within the following 12 hours. Therefore, the failure to identify loss of control rod position indication and thus take the required TS actions is a condition prohibited by TS and is reportable under 10 CFR 50.73(a)(2)(i)(B).

III. Analysis of the Event

On 3/16/17, Limerick Unit 1 MCR received an alarm for "RPIS INOPERATIVE" due to a power supply failure. Review of indicators by the control room staff found that several indicators were impacted and did not show control rod position for some of the control rods.

The operating crew discussed the observed indications and impacts of the power supply failure. Since the majority of the impacted control rods were either full-in or full-out, it was concluded that the red or green indicator on the Full Core Display satisfied the RPIS rod position TS action for those control rods. This left two control rods without position indication. Rod position was obtained using an alternate method, which met the TS required action.

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Following restoration of operability, it was identified that the failure of the power supply rendered the full core display incapable of updating in response to a position change for the affected control rods. An additional 81 control rods had no position indication while the power supply was failed.

Therefore, it was determined that the TS required actions for the control rods had not been met following the power supply failure. TS 3.1.3.7 provides several options to either determine control rod position or insert the control rod within 1 hour after discovery. If one of these options cannot be met within 1 hour, an action is entered to be in hot shutdown within the following 12 hours. The time between the initial power supply failure and replacement of the power supply was approximately 19.5 hours and exceeded the total 13 hours to complete the TS required shutdown.

Following restoration of RPIS indication it was confirmed that the control rods had remained at their original positions. Per TS Bases, the RPIS system must be operable in order to ensure that the control rod pattern can be followed and therefore other parameters maintained within their limits. Per the Updated Final Safety Analysis Report (UFSAR) Section 7.7.1.2, the Reactor Manual Control System and RPIS Subsystem are power generation systems and not safety related, nor required to fulfill a plant safety function associated with any design basis events or transients. The RPIS system is not included in the Probabilistic Risk Analysis (PRA) program. There was no control rod movement that occurred and none was required during the time that rod position was inoperable. There was no safety consequence for this event.

IV. Cause of the Event

A Corrective Action Program Evaluation (CAPE) was performed to determine the cause of the failure to identify a loss of control rod position. The evaluation concluded that procedural guidance was inadequate in identifying what indicators are available given failure of the power supply.

The lack of proceduralized guidance on which RPIS indicators are available to be used for RPIS operability and limitations on their use is the primary cause of this event. The procedures provide an incomplete list of indicators and no cautions or limitations on the "memory" within RPIS that could cause false indication. During this event, the procedure provided options to perform this via the Control Rod Position Report (preferred) or the Four Rod Display. The other potential indicators that provide control rod position were not addressed.

Based on this evaluation, it was determined that the cause of this event is inadequate procedural guidance regarding control rod position indicators available and limitations on their use. This includes which indicators can be used to satisfy surveillance requirements and how to confirm that the indicated position is updating with current data. A contributor to this event is that Operations personnel were not aware that some RPIS indicators are based on system memory.



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V. Corrective Actions Completed/Planned

The Operations procedures for RPIS indication will be revised to address all available indicators and methods to verify the indicators are accurate. The training program will be updated with information on the memory used for control rod position indication.

VI. Previous Similar Occurrences

On 5/27/2006, a Unit 2 power supply failed causing a loss of rod position indication on the Plant Monitoring System, the Four Rod Display, and the Rod Worth Minimizer (RWM). The Full Core Display was initially used to verify all rods full-in or full-out. Following review of the vendor manual, it was determined that the Full Core Display was showing position from memory and not current position based on the failure. The power supply was replaced within the LCO action window. This event had similar action taken by the operators in use of the Full Core Display. This was a lost opportunity to revise procedures to document the findings regarding the Full Core Display indication.